

AP CSP | Lesson 09: Expressions & Strings

Teacher answer key and marking guide

30 teacher marks. Original AP-style practice; this worksheet does not predict an AP score.

Award each written-response point independently. Accept equivalent correct wording and valid alternative methods. For Q5, award its one mark only for both correct choices and no others.

Questions 1-6 | 1 mark each

Q1. Answer: A

A is correct: y received a copy of 6. B treats assignment as a live link. C invents addition. D confuses a stored result with an automatically recalculated formula.

Q2. Answer: B

B is correct: multiply 3 by 5, then add 4. A adds before multiplying. C and D do not follow the given operations and their precedence.

Q3. Answer: C

C is correct: / performs division, so the result can be decimal. A truncates incorrectly. B is $13 \text{ MOD } 4$. D is not the quotient.

Q4. Answer: D

D is correct: $23 = 3 \times 6 + 5$. A is the number of complete groups. B is the division result. C equals the divisor and cannot be this remainder.

Q5. Answer: B and C

B takes three characters at positions 2, 3 and 4. C has no remainder. A is false because CONCAT adds no space; it produces "AB". D is false because $9 / 2$ is 4.5.

Q6. Answer: A

A selects two characters, at positions 3 and 4. B starts at position 2. C takes three characters. D starts at position 1. The third argument is a count, not an end position.

Written-response answers | Questions 7-12

Q7. 4 marks

Answer: (a) 18: multiplication first. (b) 30: parentheses first. (c) 3.75. (d) 6: MOD first, then addition.

1 mark: (a) 18, with $4 * 3$ evaluated first.

1 mark: (b) 30, with $6 + 4$ evaluated first.

1 mark: (c) 3.75 from division.

1 mark: (d) 6, with $15 \text{ MOD } 4 = 3$ evaluated before addition.

Q8. 4 marks

Answer: (a) 3 packs. (b) 5 tiles. (c) $26 = 3 \times 7 + 5$. (d) 0, 1, 2, 3, 4, 5, 6.

1 mark: 3 full packs.

1 mark: Remainder 5.

1 mark: $26 = 3 \times 7 + 5$ (or equivalent equality).

1 mark: Exactly the integers 0 through 6.

Q9. 4 marks

Answer: (a) "RIS". (b) "RIS!". (c) Positions 4, 5 and 6. (d) "SUNRISE".

1 mark: part is "RIS".

1 mark: tag is "RIS!".

1 mark: Positions 4, 5 and 6, demonstrating a three-character count.

1 mark: s remains "SUNRISE"; neither later assignment targets s.

Q10. 4 marks

Answer: (a) 1.25. (b) 120. (c) hours stores the earlier evaluated result; it is not assigned again. (d) $\text{hours} \leftarrow \text{minutes} / 60$.

1 mark: Final hours is 1.25.

1 mark: Final minutes is 120.

1 mark: Explains that the earlier result is stored and no later assignment updates hours.

1 mark: $\text{hours} \leftarrow \text{minutes} / 60$ (equivalent valid calculation accepted).

Q11. 4 marks

Answer: "Hello, Zoe"; "Zo"; 2.5; 30.

1 mark: Greeting is "Hello, Zoe", with the supplied space.

1 mark: Badge is "Zo" with case preserved.

1 mark: Decimal hours is 2.5.

1 mark: Remaining minutes is 30.

Q12. 4 marks

Answer: (a) $\text{badge} \leftarrow \text{SUBSTRING}(\text{name}, 1, 2)$. (b) Positions start at 1. (c) For example, 30 minutes. (d) 0.5 hours; this distinguishes decimal division from an incorrect result of 0.

1 mark: Correct assignment uses start 1 and length 2.

1 mark: States the supplied procedure uses 1-based positions.

1 mark: Chooses an integer minute value from 1 through 59.

1 mark: Gives the correct fractional hours for that choice and explains that it reveals improper truncation.